INCH-POUND

MIL-P-50570A (AR)
AMENDMENT 7
31 July 1989
SUPERSEDING
AMENDMENT 6
10 October 1986

MILITARY SPECIFICATION

PROJECTILE, 155MM, HE, ROCKET ASSISTED
M549 & M549A1, DELAY ASSEMBLY FOR

This amendment forms a part of Military Specification MIL-P-50570A (AR), dated 19 July, 1979, and is approved for use within the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

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2.1: Delete the following Military Specification:

"MIL-E-463 - Ethyl Alcohol, For Ordnance Use"

Add the following Military Standards:

"MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification

MIL-STD-00453 - Inspection, Radiographic"

Delete all the Inspection Equipment Drawings with the exception of 9246401 and 9246420. Add the following new inspection equipment drawing:

"9235999 PJCTL, 155MM, HE, RA Assembly, M549".

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3.3.1: Delete in its entirety.

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3.4: Flight test. Delete in its entirety and substitute the following:

"3.4 Flight test.

3.4.1 Delay chemicals acceptance.

3.4.1.1 Pilot mixtures. Small batches from each new lot of chemicals shall be prepared as shown in Table II.

TABLE II. Pilot mixtures.

Batch #	%W	%$BaCrO_4$	%$KClO_4$
1	45.00 ± .10	40.50 ± .10	14.50 ± .10
2	46.00 ± .10	39.50 ± .10	14.50 ± .10
3	47.00 ± .10	38.50 ± .10	14.50 ± .10
4	48.00 ± .10	37.50 ± .10	14.50 ± .10
5	49.00 ± .10	36.50 ± .10	14.50 ± .10

% VAAR shall be kept constant at approx. 1%.

3.4.1.2 Testing. Ten delay assemblies from each batch shall be submitted for ballistic firing. After conditioning at 70°F for a period of 24 hours minimum, the delays shall be flight tested in accordance with 4.5.1.1.1.

3.4.1.3 Tungsten curve. The ballistic results shall be plotted against the percent tungsten. From this curve the optimum tungsten percent will be determined. The chemical composition is optimized at a delay time of 7.0 seconds.

3.4.1.4 Usage. The results from paragraphs 3.4.1.1 - 3.4.1.3 shall be used in subsequent delay composition preparation.

3.4.2 Delay composition acceptance.

3.4.2.1 Plus 70°F. After conditioning at 70°F for a period of 24 hours minimum (min), the delay assembly shall be flight tested at a proving ground as specified in 4.4.3.3.1.1. The average delay time shall be 7.75 seconds (sec) maximum and 6.25 sec min. The max. allowable standard deviation is 0.5 sec.

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3.4.2.2 Plus 145°F. After conditioning at +145°F for a period of 24 hours min., the delay shall be flight tested at a proving ground as specified in 4.4.3.3.1.2. The average delay time shall be 5.60 sec. min., and 7.10 sec. max. The max. allowable standard deviation is 0.5 sec.

3.4.3 Delay assembly acceptance.

3.4.3.1 Plus 145°F. After conditioning at 145°F for a period of 24 hours minimum, the delay assembly shall be flight tested at a proving ground (see 4.4.3.3.2.1). The average delay time shall be 6.35 seconds plus or minus .75 seconds, and the standard deviation shall be no greater than .8 seconds.

* 3.4.3.2 Minus 65°F. After conditioning at minus 65° for a period of 24 hours minimum, the delay assemblies shall be flight tested at a proving ground (see 4.4.3.3.2.2). The average delay time shall be 8.50 seconds plus or minus 1.5 seconds, and the standard deviation shall be no greater than .8 seconds.

3.4.3.3 Plus 70°F. After conditioning at 70°F for a period of 24 hours min., the delay assembly shall be flight tested at a proving ground (see 4.4.3.3.2.3). The average delay time shall be 7.90 sec. max. and 6.10 sec. min. The max. allowable standard deviation is 0.5 sec."

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3.5: Delete in its entirety and substitute the following:

"3.5 Crimp strength. The washer crimp shall be capable of withstanding a 225 pound minimum load when tested as specified in 4.4.3.4."

Add the following new paragraph:

"3.10 Process control for delay increment consolidation. A process control description shall be developed for delay increment consolidation and submitted for government approval. The description shall include all elements of press setup, operation, calibration, maintenance, and operator instruction required to assure that the required number of delay increments are individually consolidated at the required pressure and for the required minimum consolidation time. The process controls shall include verification that the required pressure and dwell are met at the extremes of the press consolidation range (see 4.4.3.1 and 6.9)."

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4.4.2.4: Delete Minor defect 201 in its entirety.

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4.4.2.5: Add the following Minor defect:

"Minor 203: "Distance from end of tube to inside bottom of igniter plug, min. 0.65% 3.2 Gage".

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4.4.3.1: Delete in its entirety and substitute the following:

"4.4.3.1 Consolidation load. Critical Defect. The contractor shall provide adequate controls to assure the compliance of each lot with the specified load requirement and dwell time. Proper calibration of each consolidation machine shall be verified at the beginning, middle and end of each production shift. Dwell time and consolidation load shall be measured with a stopwatch and a load cell. If any consolidation machine fails verification for dwell time, consolidation load or for not maintaining proper load through the dwell time, all the delays manufactured on that machine since the last acceptable verification shall be rejected. In addition, a warning device indicating failure of dwell time and failure of consolidation load shall be assembled to each consolidation machine. The warning device or mechanism shall provide an easily detected signal to the machine operator. If during loading of the delay the consolidation load or dwell time requirements are not met, that delay shall be rejected."

4.4.3.3.1.1: Change reference paragraph 3.4.1.1 to 3.4.2.1.

* 4.4.3.3.1.1: Delete in its entirety and substitute the following:

"4.4.3.3.1.1 Plus 70°F. Each lot of delay composition shall be equally divided into ten sub-lots. One delay assembly shall be loaded from each sub-lot. The ten sample delay assemblies shall be submitted to the Government for flight testing. If the requirement of paragraph 3.4.2.1 is not met - Major defect, or if the delay assembly fails to ignite - Major defect, or ignites in the tube - Critical defect, the delay assembly lot shall be rejected. The test shall be conducted in accordance with paragraph 4.5.1.1.1. A minimum of 8 radar delay times shall be required to determine delay time acceptance."

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4.4.3.3.1.2: Change reference paragraph 3.4.1.2 to 3.4.2.2.

* 4.4.3.3.1.2: Delete in its entirety and substitute the following:

"4.4.3.3.1.2 Plus 145°F. Each lot of delay composition shall be equally divided into ten sub-lots. One delay assembly shall be loaded from each sub-lot. The ten sample delay assemblies shall be submitted to the government for flight testing. If the sample fails to meet the requirements of 3.4.2.2 - Major defect, or if any delay functions in the tube-critical defect, or if any delay fails to function-Major defect, the lot shall be rejected. The test shall be conducted in accordance with 4.5.1.1.2. A minimum of 8 radar delay times shall be required to determine delay time acceptance."

4.4.3.3.2.1: Change reference paragraph 3.4.2.1 to 3.4.3.1.

* 4.4.3.3.2.1: Delete in its entirety and substitute the following:

"4.4.3.3.2.1 Plus 145°F. A sample of ten (10) loaded delay assemblies shall be selected at random and forwarded to a Government proving ground for firing. If the sample fails to meet the requirements of 3.4.3.1 - Major defect, or if any delay functions in bore - Critical defect, or if any delay fails to function - Major defect, the lot shall be rejected. The test shall be conducted in accordance with 4.5.1.2.1. A minimum of 8 radar delay times shall be required to determine delay time acceptance."

4.4.3.3.2.2: Change reference paragraph 3.4.2.2 to 3.4.3.2.

* 4.4.3.3.2.2: Delete in its entirety and substitute the following:

"4.4.3.3.2.2 Minus 65°F. Commencing with the first production lot and then with every third lot thereafter, a sample of five (5) loaded delay assemblies shall be selected at random and forwarded to a Government proving ground for firing. If the sample fails to meet the requirements of 3.4.3.2 - Major defect, or if any delay functions in bore - Critical defect, or if any delay fails to function - Major defect, the lot shall be rejected. The test shall be conducted in accordance with 4.5.1.2.2. A minimum of 4 radar delay times shall be required to determine delay time acceptance."

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4.4.3.3.2.3: Change reference paragraph 3.4.2.3 to 3.4.3.3.

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4.4.3.8: Add the following at the end of the paragraph:

"Should the solids content be too high, additional solvent may be added, with agitation, until the solids content is acceptable. The VAAR solution will then be retested to assure compliance."

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Add the following new paragraph:

"4.4.3.9 Density of delay composition pellets. A sample of 5 pellets shall be randomly selected from each lot and the density determined in accordance with 4.5.5 for compliance in accordance with the requirements on Drawing 9235983. If any pellet does not meet the required density, the lot shall be rejected."

4.5.1.1.1: Delete the first paragraph and substitute the following:

"Plus 70°F. The loaded delay assemblies shall be pressed into M549 projectiles in accordance with drawing 9235999. The M549 projectiles shall have live rocket grains and inert filled warheads. To assure proper pressing of the delays the assembled test projectiles shall be radiographed."

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4.5.1.1.2: Delete in its entirety and substitute the following paragraph:

"4.5.1.1.2 Plus 145°F. This test shall be conducted as specified in 4.5.1.1.1 except that the conditioning temperature for the delay assembly and projectile shall be plus 145°F and a PXR-6297 proof charge conditioned at +145°F is to be used in lieu of a M203 series propelling charge."

4.5.1.2.1: Delete in its entirety and substitute the following paragraph:

"4.5.1.2.1 Plus 145°F. This test shall be conducted as specified in 4.5.1.1.1 except that the conditioning temperature for the delay assembly and projectile shall be plus 145°F and a PXR-6297 proof charge conditioned at plus 145°F is to be used in lieu of a M203 series propelling charge."

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4.5.3: Delete the last sentence and add the following at the end of the paragraph:

"In lieu of a penitrometer, a radiographic standard shall be manufactured to assure adequate sensitivity of the radiographic technique. A delay will be manufactured with a low density pocket in each of the two delay columns. Normal production materials shall be used to create the standard delay. In both columns the flash and first delay increment shall be pressed in correctly. In one column the second and third delay increments shall be placed in the column but not pressed, but the fourth and fifth increments shall be pressed. In the other column the second increment will be correctly pressed, but the third and fourth increments will be loose loaded, the fifth increment will be pressed. Both columns will be back drilled, have igniter pressed in and have washers assembled. The delay manufactured as above will have a void between the first and second delay increments in one column and a void between the second and third increments in the other column. These voids must be clearly visible when the standard delay is radiographed. Using the standard delay, film sensitivity shall be checked on each piece of film used for delay acceptance. Photographic density of the film used shall be no less than 1.25 as determined in accordance with 4.5.4."

Add the following new subparagraph:

"4.5.3.1 Operator qualification. For operation of radiograph equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of radiograph equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II."

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Add the following new paragraph:

"4.5.5 Determination of density. The density of the delay composition pellet shall be calculated as follows:

$$\text{Density} = \frac{\text{Pellet wt.}}{(\text{Length of Pellet}) \quad (\text{Dia. of Pellet}) \quad (.7854)}$$

Note: All weights and dimensions are to be used in metric units."

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6.6: Delete in its entirety and substitute the following:

6.6 Pilot mixture. Composition for use in M549 delays may also be qualified by the pilot mixture tests prescribed in MIL-P-63140 (M650 Delay). This one pilot mixture test may qualify composition for both M549 and M650 delay assemblies.

Add the following new paragraph:

"6.9 Submission of process control description. The process control description for delay increment consolidation shall be prepared in accordance with DI-E-10392 and submitted to the responsible government technical agency for approval.

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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